22/07

Expt 01: Study of various Network commands used in Linux and Window.

arp -a

ARP is a short form of addres resolution protocol. It shows ip addres of your computer and MAC addres of your router.

Op: - gateway (10.0.2.2) at 52.54:00:12:35:02 [ether] on eposs

hostname: displays the name of your compiler

Op: localhost.localdomain

ipconfig /all: displays detailed network configuration for all network adapter on your computer

Window Ip Configuration

Hostname: Vishnu Priya

Primary Dns System ....:

Node Type ..... Mixed

Ip routing enabled ....: No

WINS proxy enabled ...: No.

nbstat -a: helps to solve problem with NetBIOS name resolution

Op: Displays protocol statistics and current TCP/IP connection using NBT

-a (adapter status) Lists the remote machine name table given its name

-A (Adapter status) List the remote machine name table given its Ip addres

netstat: network statistics

Op:

Proto	Local Addres	foreignAddres	State
TCP	10.232.78.95:49408	4.213.25.242:http	ESTABLISHED
TCP	10.232.78.95:49410	4.213.25.242:https	ESTABLISHED

nslookup: It is a tool used to perform DNS lookup in Linux

Op: Default Server: Unknown

Addres : 10.232.78.50

pathping : The path it take to reach destination and where network fails in between.

op: -g host-list loose source route along host-list
-h maximum_hops Maximum number of hops to search for target

ping: Packet INternet Groper.
helps to check connectivity between two points

op: -t ping the specified host until stopped.
To see statistics and continue - type control-Break;

route: route command is used to show / manipulate Ip routing table.

op: -f clears the routing table of all gateway entries. If this is used in conjunction with one of commands, the tables are cleared prior to running the command.

Linux Network commands

ip: It is used to troubleshooting existing systems. The ip command can show address information, manipulate routing, plus display network various devices, interfaces and Tunnels

ip <OPTION> <OBJECT> <COMMAND>

(a) to show ip address assigned to an interface on your server

ip address show

1:lo: <LOOPBACK, up, LOWER_UP> mtu 65536

qdisc noqueue stat unknown group default qlen 1000.

(b) ip address add 192.168.1.254/24 dev enp0

(c) ip address del 192.168.1.254/24 dev enp3

2) ifconfig : It was a staple in many sysadmin tool belt for configuration and troubleshooting between network.

op: enp0s3 : flags 0183 <UP, BROADCAST, RUNNING, MULTICAST>
mtu 1500 inet 10.0.2.15 netmask 255.255.255.0 ....

3) mtr: MTR is a program with a command line interface that serves as a network diagnostic and troubleshooting tool. This command combines the functionality of the ping and traceroot command

mtr

op: my traceroute [v0.94]

	host	Packets		Pings				
1	localhost	Loss%	snt	Last	Avg	Best	wrst	stDev
		0.0%	122	0.1	0.2	0.1	1.9	0.2

(a) #mtr google.com

op: host
1. gateway
2. (waiting for reply)

(b) #mtr -g google.com

op: -f filename FILE
-u --tcp

4) tcpdump : The tcpdump command is designed for capturing and displaying packet.

op:
tcpdump : enp0s3 : you don't have permission to capture on that device

5) ping
ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Message.

Command : ping google.com

OP: PING google.com (142.200.182.78) 56(84) bytes of data 64 byte from demaa-ax-in-f4.1e100.net (142.250.182.78): icmp - seq =1 ttl =112 time =9.38 ms